Abstract

The increasing adoption of generative artificial intelligence (GenAI) in higher education has raised concerns about students' reliance on AI-assisted technologies and their perceived effects on independent problem-solving. This study investigated the association between GenAI dependency and perceived negative effects on problem-solving skills or efficiency among university students. A quantitative, cross-sectional research design was employed using secondary survey data from 227 active GenAI users in the United States and Indonesia. A seven-item dependency composite was constructed from Cognitive Preoccupation and Withdrawal measures, excluding Negative Consequences items to reduce direct measurement overlap with the outcome. Data were analyzed using descriptive statistics, reliability assessment, Spearman's rank-order correlation, ordinal logistic regression, and sensitivity analyses. The dependency composite demonstrated good internal consistency (Cronbach's α = .801). Spearman's correlation identified a significant positive association between dependency and perceived negative problem-solving impact (ρ = .387, $p < .001$). The primary adjusted ordinal logistic regression showed that higher dependency scores were associated with greater odds of reporting negative perceived problem-solving effects (adjusted OR = 2.946, 95% CI [2.065, 4.204], $p < .001$). Withdrawal remained significantly associated with the outcome when examined alongside Cognitive Preoccupation. Exploratory analysis found no statistically significant evidence that the dependency association differed between respondents from the United States and Indonesia (p = .596). The findings suggest that stronger GenAI dependency-related experiences are associated with greater perceived negative problem-solving impact. However, the cross-sectional design and self-reported measurements prevent causal conclusions or claims of objectively measured cognitive decline. The study highlights the importance of responsible GenAI integration and further research using longitudinal designs and objective problem-solving assessments.

Keywords: generative artificial intelligence, AI dependency, problem-solving, university students, ordinal logistic regression, higher education

Generative AI Dependency and Perceived Negative Effects on Problem-Solving Among University Students: A Cross-Sectional Study

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1. Introduction

1.1 Background and Motivation

Generative artificial intelligence (GenAI) has become an increasingly important technological resource in higher education, influencing how university students access information, complete academic assignments, and approach complex learning activities. Tools such as ChatGPT provide conversational assistance, explanations, content generation, and problem-solving support, creating opportunities for students to improve their learning efficiency. However, the growing integration of these technologies into academic activities has also raised concerns about excessive reliance and its potential relationship with students' independent cognitive functioning.

Problem-solving is an essential higher-order cognitive skill that enables students to identify challenges, evaluate available information, develop appropriate strategies, and make informed decisions. Within university education, the development of this ability is particularly important because academic success requires more than obtaining correct answers. Students must also demonstrate independent reasoning, analytical judgment, and the ability to apply knowledge in unfamiliar situations.

Existing research suggests that the educational consequences of GenAI are complex. Chen and Cheung (2025), in a systematic review and meta-analysis, reported positive effects of GenAI on several university learning outcomes, including higher-order thinking. Similarly, Daniel et al. (2025) identified evidence that GenAI-supported learning can contribute to the development of cognitive and academic skills. These findings demonstrate that AI-assisted learning may offer substantial educational benefits when implemented appropriately.

Nevertheless, concerns remain regarding the extent to which students may become dependent on AI-generated assistance. Goh et al. (2025) conceptualized generative AI dependency as a multidimensional construct involving cognitive preoccupation, negative consequences, and withdrawal. Their research demonstrated that dependency-related experiences are associated with several psychological, motivational, and behavioral outcomes, including indicators related to critical thinking and task performance.

These findings suggest that the central educational concern may not simply be whether students use AI, but how they engage with it and whether reliance on AI becomes associated with perceived difficulties in independent cognitive activities.

1.2 Research Gap

Although the literature increasingly examines the educational benefits and risks of GenAI, important questions remain concerning the relationship between dependency-related experiences and students' perceptions of their problem-solving capabilities. Research on AI-assisted learning frequently emphasizes academic achievement, learning efficiency, or general critical-thinking outcomes, while dependency research often considers broader psychological and behavioral consequences.

A further methodological concern arises when researchers use overlapping questionnaire items to measure both AI dependency and its proposed consequences. Such overlap may artificially strengthen observed associations and complicate interpretation.

The present study addresses this measurement concern by constructing a seven-item dependency composite that excludes the Negative Consequences dimension, including the item used to assess perceived negative effects on problem-solving. This approach enables a clearer examination of dependency-related experiences and the selected outcome, although both remain self-reported measures from the same questionnaire.

1.3 Research Purpose and Contributions

The primary purpose of this study is to examine the association between generative AI dependency and university students' perceived negative effects of AI use on their problem-solving skills or efficiency.

Using secondary survey data from 227 active GenAI users in the United States and Indonesia, the study applies descriptive statistics, reliability analysis, Spearman correlation, and ordinal logistic regression to investigate this relationship.

The study makes three principal contributions. First, it provides an empirical examination of the association between dependency-related experiences and perceived negative problem-solving impact using a five-category ordinal outcome. Second, it applies a study-specific dependency measure designed to reduce direct item overlap between the predictor and outcome. Third, it evaluates the robustness of the primary association through alternative model specifications, dimension-specific analyses, and exploratory cross-country comparisons.

The findings may inform discussions about responsible GenAI integration in higher education and provide a basis for further investigation using longitudinal designs and objective assessments of problem-solving performance.

1.4 Scope of the Study

The study uses an existing cross-sectional survey dataset originally collected from undergraduate business students in the United States and Indonesia (Pertiwi & Adiningrum, 2026). The analysis focuses on 227 respondents identified as active GenAI users.
The outcome is a self-reported assessment of whether GenAI use has negatively affected problem-solving skills or efficiency. Consequently, the study examines perceived negative impact rather than objectively measured problem-solving ability.
Because the data are observational and cross-sectional, the findings cannot establish causal effects or determine whether AI dependency preceded changes in students' perceived problem-solving abilities. Furthermore, the sample is limited to the populations represented in the original survey, restricting generalizability to university students more broadly.

2. Related Work and Literature Review

2.1 Generative Artificial Intelligence in Higher Education

The growing adoption of generative artificial intelligence (GenAI) has transformed how university students access academic information, complete assignments, and engage in independent learning. Unlike traditional information retrieval systems, GenAI applications can generate explanations, summarize complex concepts, provide feedback, and assist with analytical tasks. These capabilities have encouraged researchers to examine both the educational benefits and potential risks of AI-assisted learning.
Kasneci et al. (2023) examined the opportunities and challenges of large language models in education, identifying potential applications in personalized instruction, feedback generation, and learning support. However, the authors also emphasized concerns relating to misinformation, algorithmic bias, and excessive reliance on AI-generated content. Their findings suggest that educational outcomes depend partly on how these technologies are incorporated into learning activities.
Similarly, Lo (2023) conducted a rapid review of ChatGPT in education and identified opportunities for instructional support, assessment assistance, and academic content development. Nevertheless, the review highlighted challenges involving academic integrity, accuracy, and the responsible use of AI-generated information.

Tlili et al. (2023) further examined ChatGPT as an educational chatbot, discussing its potential to improve access to learning resources while identifying risks associated with unreliable information and ethical concerns. Collectively, these studies demonstrate that GenAI offers considerable educational opportunities but requires appropriate safeguards to support meaningful learning.

2.2 Generative AI, Critical Thinking, and Problem-Solving

Critical thinking and problem-solving are fundamental competencies in higher education. They involve evaluating evidence, identifying relationships, assessing alternative solutions, and making reasoned decisions. Although generative artificial intelligence (GenAI) can support these activities, concerns have emerged regarding whether excessive dependence on AI-generated answers may reduce opportunities for independent cognitive engagement.
Chen and Cheung (2025) examined the effects of GenAI on university students' learning outcomes through a systematic review and meta-analysis. Their findings suggested that GenAI-supported learning can improve several educational outcomes, including aspects of higher-order thinking. This indicates that AI use does not necessarily undermine cognitive development and may offer educational benefits when appropriately integrated into learning activities.
Tian and Zhang (2025) investigated the relationship between AI dependence and critical thinking among 580 university students in China. Their findings identified a negative association between AI dependence and critical thinking, with cognitive fatigue partially mediating this relationship. The researchers also reported that information literacy influenced the relationships among AI dependence, cognitive fatigue, and critical thinking. These findings suggest that psychological experiences and students' ability to evaluate information may be relevant when examining dependency-related academic outcomes.
Similarly, Hou et al. (2025) investigated undergraduate students' reliance behaviors during GenAI-supported problem-solving. Their research distinguished reflective, cautious, thoughtless, and collaborative patterns of AI reliance. The findings indicated that critical-thinking skills and dispositions were positively associated with reflective, cautious, and collaborative reliance behaviors. This suggests that the educational implications of AI use may depend on how students interact with AI-generated assistance rather than simply whether they use these technologies.

Goh et al. (2025) further examined GenAI dependency and its motivational, behavioral, and psychological correlates. Their work is particularly relevant because it conceptualized dependency as a multidimensional phenomenon rather than treating all AI use as equivalent. Collectively, these studies highlight the importance of distinguishing productive AI assistance from dependency-related experiences. Students who use GenAI to obtain explanations, critically evaluate suggestions, and develop independent solutions may experience different academic outcomes from those who rely on AI-generated responses without sufficient evaluation.

However, the existing evidence does not establish that GenAI dependency necessarily causes deterioration in problem-solving ability. Further investigation is required to distinguish perceived negative effects from objectively measured cognitive outcomes.

2.3 Generative AI Dependency and Its Dimensions

Generative AI dependency refers to patterns of psychological or behavioral reliance on AI systems. Such reliance may involve persistent thoughts about AI use, discomfort when access is unavailable, or perceived negative consequences associated with continued use.

Goh et al. (2025) developed and validated a Generative AI Dependency Scale comprising three dimensions: Cognitive Preoccupation, Negative Consequences, and Withdrawal. Cognitive Preoccupation reflects persistent attention toward AI use, including the desire to engage with AI even when it is unnecessary. Withdrawal captures experiences such as restlessness, frustration, or discomfort when AI tools are unavailable. Negative Consequences concerns perceived difficulties associated with reliance on GenAI.

A related measurement perspective was presented by Hou et al. (2025), who developed and validated a scale assessing undergraduate students' reliance behaviors during GenAI-supported problem-solving. Their analysis identified four dimensions: reflective use, cautious use, thoughtless use, and collaborative use. Unlike measures emphasizing dependency-related psychological experiences, this instrument focused on students' behavioral approaches to AI-assisted problem-solving.

These measurement frameworks highlight an important conceptual distinction. Psychological dependency and behavioral reliance are related research areas, but they should not automatically be treated as identical constructs. A student may frequently use AI while critically evaluating its responses, whereas another may experience discomfort when AI assistance is unavailable.

The multidimensional framework proposed by Goh et al. (2025) provides the foundation for the present study because the secondary dataset contains items derived from this dependency instrument.

However, the distinction between dependency and its consequences creates a measurement challenge. If a questionnaire item describing reduced problem-solving ability contributes to both the dependency predictor and the outcome, the resulting association may partly reflect shared measurement content.

The present study addresses this concern by excluding the Negative Consequences dimension from its primary dependency composite. Instead, Cognitive Preoccupation and Withdrawal items are combined into a study-specific seven-item measure, while perceived negative problem-solving impact is assessed separately.

This approach reduces direct item overlap, although it does not eliminate the possibility of common-method bias or establish that the modified composite has the same psychometric properties as the original scale.

2.4 Cross-Cultural Perspectives on AI Dependency

The relationship between GenAI use and academic experiences may vary across educational and cultural contexts. Differences in instructional practices, technological environments, institutional expectations, and students' attitudes toward AI may influence how dependency-related experiences are reported.

Pertiwi and Adiningrum (2026) provided a cross-cultural dataset examining generative AI dependency among business students in the United States and Indonesia. The dataset includes responses relating to AI dependency, critical thinking, task performance, and students' experiences with generative AI technologies.

The availability of respondents from two national contexts provides an opportunity to examine whether the association between AI dependency and perceived negative problem-solving impact differs across these groups.

Nevertheless, cross-country comparisons require careful interpretation. Observed differences may reflect variation in sample composition, educational environments, response styles, or the interpretation of questionnaire items. Without formal evidence of measurement equivalence, national comparisons should be considered exploratory rather than definitive evidence of cultural differences.

2.5 Methodological Limitations in Existing Research

Research on the educational consequences of GenAI faces several methodological challenges.

First, studies based on cross-sectional surveys cannot establish temporal ordering or causal relationships. An association between AI dependency and perceived difficulties in problem-solving does not demonstrate that dependency caused those difficulties.

Second, self-reported assessments of cognitive functioning may differ from objectively measured performance. Students' perceptions of declining problem-solving ability may be influenced by personal expectations, confidence, or other unmeasured characteristics.

Third, measurement overlap can complicate interpretation when predictors and outcomes contain similar questionnaire content. Separating dependency-related experiences from items describing negative consequences can improve conceptual clarity, although further validation remains necessary.

Finally, statistical models should reflect the measurement properties of the outcome variable. When responses are recorded on an ordered five-point agreement scale, ordinal regression provides an approach that preserves the ranking of responses without assuming equal distances between categories.

2.6 Identified Research Gap

The reviewed literature establishes that GenAI can support higher education while also raising concerns about excessive reliance. However, the relationship between dependency-related experiences and students' perceived negative problem-solving impact requires careful investigation, particularly when predictor and outcome items originate from the same dependency instrument.

The present study contributes to this discussion by examining a study-specific dependency composite that excludes directly overlapping Negative Consequences items. Using secondary survey data from university students in the United States and Indonesia, the analysis investigates the association through ordinal logistic regression and evaluates its robustness using alternative specifications and dimension-specific analyses.

Rather than assuming that AI dependency causes reduced problem-solving ability, the study focuses on the extent to which dependency-related experiences are associated with students' perceptions of negative effects on their problem-solving skills or efficiency.

3. Problem Statement

The increasing integration of generative artificial intelligence (GenAI) into university education has created opportunities for students to access information, complete academic tasks, and receive immediate learning support. However, concerns remain regarding the extent to which reliance on these technologies may be associated with students' independent cognitive functioning, particularly their perceived problem-solving skills and efficiency. Although previous research has examined the educational benefits and psychological implications of GenAI use, an important methodological challenge arises when AI dependency and its consequences are measured using overlapping questionnaire items. Such overlap can complicate the interpretation of statistical associations because the predictor and outcome may partly represent the same underlying experiences. Furthermore, research based on self-reported measures cannot establish whether students' actual problem-solving abilities have improved or deteriorated.

The present study addresses these concerns by examining the association between GenAI dependency-related experiences and students' perceived negative effects of AI use on problem-solving. Specifically, it constructs a seven-item dependency composite using Cognitive Preoccupation and Withdrawal items while excluding the Negative Consequences dimension containing the problem-solving outcome.

Using secondary survey data from 227 university students in the United States and Indonesia who reported active GenAI use, the study employs ordinal logistic regression to evaluate the association while adjusting for selected demographic and AI-use characteristics. The analysis also investigates the separate associations of Cognitive Preoccupation and Withdrawal and explores whether the primary relationship differs between the two national groups.

The study seeks to provide statistically grounded evidence about perceived problem-solving consequences associated with GenAI dependency while acknowledging the limitations of cross-sectional, self-reported data.

4. Research Questions and Hypotheses

4.1 Research Questions

RQ1: What is the relationship between generative AI dependency and students' perceived negative effects of AI use on their problem-solving skills or efficiency?

RQ2: Does the association between generative AI dependency and perceived negative problem-solving impact remain statistically significant after adjusting for country, academic year, and frequency of AI use?
RQ3: How do the Cognitive Preoccupation and Withdrawal dimensions of generative AI dependency relate to perceived negative problem-solving impact?
RQ4: Does the association between generative AI dependency and perceived negative problem-solving impact differ between university students in the United States and Indonesia?

4.2 Research Hypotheses

H1: Higher generative AI dependency scores are associated with greater odds of reporting a more negative perceived effect of AI use on problem-solving skills or efficiency.
H2: The positive association between generative AI dependency and perceived negative problem-solving impact remains statistically significant after adjustment for country, academic year, and frequency of AI use.
H3: Cognitive Preoccupation and Withdrawal are positively associated with perceived negative problem-solving impact when examined separately.
H4: The strength of the association between generative AI dependency and perceived negative problem-solving impact differs between students in the United States and Indonesia.
These hypotheses are evaluated using ordinal logistic regression, dimension-specific sensitivity analyses, and an exploratory country-by-dependency interaction model.

5. Research Objectives and Contributions

5.1 Research Objectives

The primary objective of this study is to investigate the association between generative AI dependency and university students' perceived negative effects of AI use on their problem-solving skills or efficiency.
The specific objectives are to:

1. Assess the distribution and internal consistency of a study-specific seven-item measure of generative AI dependency.
2. Examine the relationship between generative AI dependency and perceived negative problem-solving impact using correlation analysis and ordinal logistic regression.

3. Determine whether the observed association persists after adjustment for country, academic year, and frequency of AI use.
4. Evaluate the separate associations of Cognitive Preoccupation and Withdrawal with perceived negative problem-solving impact.
5. Explore whether the association between generative AI dependency and perceived negative problem-solving impact differs between students in the United States and Indonesia.

5.2 Research Contributions

Measurement contribution: The study constructs and evaluates a seven-item dependency composite that excludes questionnaire items describing negative consequences, thereby reducing direct measurement overlap between the primary predictor and outcome.
Empirical contribution: The study provides quantitative evidence regarding the association between dependency-related experiences and perceived negative problem-solving impact among active GenAI users.
Methodological contribution: The research applies ordinal logistic regression, alternative model specifications, reliability diagnostics, exploratory factor analysis, and sensitivity analyses to examine the robustness of the findings.
Cross-cultural contribution: The study explores the association across respondents from the United States and Indonesia while explicitly recognizing the limitations of cross-cultural comparisons without established measurement equivalence.
Educational contribution: The findings provide a basis for further investigation into responsible GenAI use and the development of educational approaches that encourage independent reasoning alongside AI-assisted learning.

6. Research Methodology

6.1 Research Design

This study adopted a quantitative, observational, cross-sectional research design using secondary survey data to investigate the association between generative artificial intelligence (GenAI) dependency and university students' perceived negative effects of AI use on problem-solving skills or efficiency. The research employed descriptive and inferential

statistical methods to examine relationships among variables without manipulating participants' exposure to AI technologies.

An explanatory association-focused approach was adopted, with ordinal logistic regression serving as the primary analytical method. This approach was appropriate because the dependent variable was measured using a five-point ordered Likert response. The analysis also incorporated reliability assessment, exploratory factor analysis, correlation analysis, sensitivity testing, and exploratory cross-country comparisons.

The study did not involve an experimental intervention or longitudinal follow-up. Consequently, the findings were interpreted as statistical associations rather than evidence of causal effects.

6.2 Data Source and Study Population

The study utilized the *When Culture Meets AI: A Cross-Cultural Study of Generative AI Dependency Among U.S. and Indonesian Business Students* dataset, published by Pertiwi and Adiningrum (2026) through Zenodo (DOI: 10.5281/zenodo.21013786).

The dataset contained survey responses from 239 participants and included demographic characteristics, generative AI usage patterns, dependency-related experiences, critical-thinking measures, and task-performance indicators. The original dataset comprised 41 variables and included a codebook documenting item descriptions, construct classifications, and response coding.

The analytical sample was restricted to respondents identified in the dataset as active GenAI users. Twelve respondents were excluded based on this eligibility criterion, resulting in a final sample of 227 participants.

Of these participants, 116 were from the United States and 111 were from Indonesia. The analysis retained the original country classifications and examined country differences as exploratory rather than representative national comparisons.

The study relied on the original survey's available participant information. No additional recruitment or primary data collection was conducted.

6.3 Data Preparation and Preprocessing

Data preparation and analysis were performed in Python using a Jupyter Notebook environment. The Excel workbook was imported, and the survey data and accompanying codebook were examined to identify relevant variables, response categories, and measurement definitions.

The preprocessing procedure involved verifying the dataset structure, applying the active-user eligibility criterion, inspecting missing values, checking participant identifiers for duplicates, and converting selected questionnaire responses into appropriate numeric formats. The analytical dataset contained 227 eligible respondents. No missing values were identified in the seven dependency items or the problem-solving outcome, and all responses to these items fell within the expected range of 1–5.

The seven-item dependency composite was independently recalculated and compared with the stored composite values. This verification identified no mismatches among the 227 respondents.

The analysis used the complete eligible sample for the primary model. No imputation, synthetic-data generation, oversampling, or class-balancing procedure was applied.

6.4 Measurement of Study Variables

6.4.1 Independent Variable: Generative AI Dependency

The primary independent variable was a study-specific seven-item AI Dependency Score derived from selected items of the Generative AI Dependency Scale described by Goh et al. (2025).

The original questionnaire organized dependency-related items into three dimensions: Cognitive Preoccupation, Negative Consequences, and Withdrawal.

To reduce direct measurement overlap between the predictor and the outcome, the present study excluded the four Negative Consequences items (GAID4–GAID7).

The final dependency composite comprised Cognitive Preoccupation items GAID1–GAID3 and Withdrawal items GAID8–GAID11.

Each item was coded on a five-point agreement scale ranging from 1 (strongly disagree) to 5 (strongly agree). The codebook indicated consistently worded items, and no reverse coding was applied.

The composite was calculated as the arithmetic mean of the seven selected items, with higher scores indicating greater endorsement of dependency-related experiences.

Internal consistency was assessed using Cronbach's alpha. The seven-item composite demonstrated good internal consistency ($\alpha = .801$). Corrected item-total correlations ranged from .404 to .653, and deletion of any individual item reduced the overall alpha.

The two theoretically defined dimensions were also examined separately. Cognitive Preoccupation was calculated from GAID1–GAID3, while Withdrawal was calculated from GAID8–GAID11.

The seven-item measure was treated as a study-specific composite rather than a separately validated version of the complete dependency instrument.

6.4.2 Dependent Variable: Perceived Negative Problem-Solving Impact

The dependent variable was measured using GAID7, which stated:

"My use of generative AI has negatively affected my problem-solving skills or efficiency."

Responses ranged from 1 (strongly disagree) to 5 (strongly agree).

Higher scores represented stronger agreement that GenAI use had negatively affected the respondent's problem-solving skills or efficiency.

The outcome was treated as an ordinal variable because the response categories were ordered, but equal distances between categories could not be assumed.

Importantly, this variable measured perceived negative problem-solving impact rather than objectively assessed problem-solving ability. Therefore, the study did not directly measure changes in cognitive performance.

6.4.3 Covariates

The primary regression model included country, academic year, and frequency of GenAI use as adjustment variables.

Country was represented by a binary indicator, with the United States serving as the reference category and Indonesia coded as 1.

Academic year was initially coded as an ordered variable representing progression from freshman to senior level.

GenAI usage frequency was coded as an ordered variable representing increasing reported frequency of use.

Alternative model specifications were used to examine the sensitivity of the primary association to covariate coding.

These variables were selected to account for measured differences in participant characteristics and AI-use patterns. However, adjustment for these variables could not eliminate the possibility of residual confounding.

6.5 Statistical Analysis

6.5.1 Descriptive Statistics

Descriptive statistics were calculated to summarize participant characteristics, AI dependency scores, and responses to the perceived negative problem-solving outcome.
Categorical variables were summarized using frequencies and percentages, while continuous composite scores were described using means, standard deviations, and distributional statistics.
The distribution of GAID7 was examined to assess the frequency of responses across the five ordered categories.

6.5.2 Correlation Analysis

Spearman's rank-order correlation was used to examine associations between the ordinal problem-solving outcome and selected dependency-related measures.
This nonparametric approach was selected because it evaluates monotonic relationships without requiring normally distributed variables.
Correlation analyses were treated as exploratory and interpreted alongside the primary regression findings.

6.5.3 Primary Ordinal Logistic Regression

The primary association was evaluated using a proportional-odds ordinal logistic regression model implemented through the OrderedModel class in Python's statsmodels library.
The model estimated the association between the seven-item AI Dependency Score and the ordered problem-solving outcome while adjusting for country, academic year, and GenAI usage frequency.
Model coefficients were exponentiated to obtain adjusted odds ratios (ORs), with 95% confidence intervals used to express estimation uncertainty.
An odds ratio greater than 1 indicated higher cumulative odds of reporting a more negative perceived problem-solving impact, under the proportional-odds parameterization used.
The model was estimated using the Broyden–Fletcher–Goldfarb–Shanno optimization algorithm, with a maximum of 1,000 iterations. Convergence was verified.
Statistical significance was evaluated using a conventional two-sided threshold of $p < .05$, while confidence intervals and effect sizes were emphasized in interpretation.

6.5.4 Model Diagnostics and Alternative Specifications

Model diagnostics included examination of convergence, multicollinearity, and the stability of the estimated dependency association.

Variance inflation factors were calculated for the primary predictors. All values were below 1.14, indicating no substantial linear multicollinearity among the included covariates.

Threshold-specific binary logistic regressions were examined as exploratory diagnostics for potential variation in predictor associations across outcome thresholds.

A multinomial logistic regression model was also estimated as an alternative specification that did not impose the same common-slope structure as the ordinal model. Model fit was compared using the Akaike Information Criterion and Bayesian Information Criterion. These comparisons were treated as supplementary diagnostics rather than a definitive formal test of the proportional-odds assumption.

6.5.5 Sensitivity and Robustness Analyses

Several sensitivity analyses were conducted to evaluate whether the primary findings depended strongly on model specification.

First, Cognitive Preoccupation and Withdrawal were examined in separate ordinal regression models and subsequently included together to assess their adjusted associations with the outcome.

Second, the dependency association was compared across unadjusted, primary adjusted, and alternative categorical-covariate specifications.

Third, a country-by-dependency interaction term was introduced to investigate whether the association differed between respondents from the United States and Indonesia. Country-specific estimates and confidence intervals were derived from the interaction model.

Fourth, a leave-one-out influence analysis was performed by repeatedly estimating the primary model after excluding one respondent at a time. This procedure assessed whether the estimated dependency association was disproportionately influenced by an individual observation.

These analyses were interpreted as robustness checks rather than independent confirmatory tests.

6.6 Exploratory Measurement Structure

Exploratory factor analysis was performed to examine the dimensional structure of the seven-item dependency composite.

Sampling adequacy was assessed using the Kaiser–Meyer–Olkin statistic, and Bartlett's test of sphericity was used to evaluate the suitability of the item correlation matrix for factor analysis.

One-factor and two-factor principal-axis solutions were estimated. The two-factor solution used oblique rotation to allow correlation between the latent dimensions.

Parallel analysis was also conducted using 1,000 simulated random datasets and a fixed random seed of 42. Observed eigenvalues were compared with the 95th-percentile eigenvalues obtained from the simulated data.

The analysis suggested one dominant factor, although the exploratory two-factor solution separated the Cognitive Preoccupation and Withdrawal items.

Because these procedures used Pearson correlations for ordinal questionnaire responses, the factor analysis was interpreted as preliminary evidence rather than comprehensive psychometric validation.

6.7 Ethical Considerations

The research involved secondary analysis of an existing survey dataset. No participants were contacted, recruited, or subjected to experimental procedures during the present analysis.

The analysis focused on the variables available in the supplied dataset and did not require the collection of additional personally identifying information.

The dataset source was acknowledged to support transparency and academic attribution.

The original study's consent arrangements, ethical approvals, and data-sharing conditions should be confirmed from its published documentation before final publication. The present study does not claim independent institutional ethics approval.

6.8 Reproducibility

The analytical workflow was implemented in Python using Jupyter Notebook. Data processing and statistical analyses employed packages including pandas, NumPy, SciPy, statsmodels, and factor-analyzer.

The notebook documented data selection, variable construction, descriptive analyses, regression estimation, sensitivity testing, and measurement diagnostics.

A fixed random seed was used for the parallel analysis to support reproducibility.
For public release, the analysis code should be accompanied by a README file, dependency specifications, the original dataset citation, and instructions for reproducing the principal results.
The publicly released materials should respect the dataset's applicable license and sharing conditions.

Code Availability

The Python code used for data preprocessing, statistical analysis, and visualization is available in the project's GitHub repository: https://github.com/abetayonoah/generative-ai-dependency-research. The original survey dataset is available through its cited Zenodo record.

7. Results and Findings

7.1 Participant Characteristics

The original dataset comprised 239 respondents, of whom 227 met the eligibility criterion of active generative artificial intelligence (GenAI) use. Twelve respondents were excluded from the analytical sample. Among the 227 eligible participants, 116 (51.1%) were from the United States and 111 (48.9%) were from Indonesia.
The sample included 137 female respondents (60.4%), 86 male respondents (37.9%), and four respondents (1.8%) who preferred not to disclose their gender. Regarding academic year, 29 participants (12.8%) were freshmen, 50 (22.0%) were sophomores, 66 (29.1%) were juniors, and 82 (36.1%) were seniors.
The largest age group was 20–22 years (n = 96, 42.3%), followed by 23–25 years (n = 68, 30.0%), 17–19 years (n = 41, 18.1%), and above 25 years (n = 22, 9.7%).
GenAI usage frequency was distributed across three categories: sometimes (n = 82, 36.1%), often (n = 108, 47.6%), and always (n = 37, 16.3%). These results indicate that most participants reported using GenAI often or always.

7.2 Descriptive Statistics of the Main Variables

The seven-item AI Dependency Score had a mean of 3.004 (SD = 0.788), with observed values ranging from 1.000 to 4.857. The median score was 3.000, indicating a moderate overall level of endorsement of dependency-related experiences within the analytical sample. The dependent variable, perceived negative problem-solving impact (GAID7), was measured on a five-point agreement scale. Forty-three respondents (18.9%) selected strongly disagree, 61 (26.9%) selected disagree, 54 (23.8%) selected neutral, 49 (21.6%) selected agree, and 20 (8.8%) selected strongly agree.

Overall, 30.4% of participants agreed or strongly agreed that GenAI use had negatively affected their problem-solving skills or efficiency. In contrast, 45.8% disagreed or strongly disagreed, while 23.8% selected the neutral response.

Table 1. Distribution of Perceived Negative Problem-Solving Impact

Response category	**Frequency**	**Percentage**
Strongly disagree	43	18.9%
Disagree	61	26.9%
Neutral	54	23.8%
Agree	49	21.6%
Strongly agree	20	8.8%
Total	**227**	**100.0%**

Mean dependency scores increased across the five outcome categories, from 2.505 among respondents who strongly disagreed with the statement to 3.550 among those who strongly agreed. This descriptive pattern suggested a positive relationship between dependency-related experiences and perceived negative problem-solving impact.

7.3 Reliability and Measurement Assessment

The seven-item AI dependency composite demonstrated good internal consistency, with Cronbach's alpha of 0.801. Corrected item-total correlations ranged from 0.404 to 0.653, and removal of any individual item reduced the overall reliability coefficient.

The Cognitive Preoccupation dimension had a Cronbach's alpha of 0.663, while the Withdrawal dimension demonstrated an alpha of 0.810.

Exploratory factor analysis indicated that the item correlation matrix was suitable for factor analysis, with a Kaiser–Meyer–Olkin value of 0.824 and a statistically significant Bartlett's test of sphericity ($\chi^2 = 446.736$, $p < .001$).

The exploratory two-factor solution showed that Cognitive Preoccupation items loaded primarily on one factor (0.709–0.874), while Withdrawal items loaded primarily on the second factor (0.789–0.802). The correlation between the two factors was 0.447.

However, parallel analysis supported retaining only one dominant factor. The first observed eigenvalue (3.219) exceeded its simulated 95th-percentile threshold (1.348), whereas the second observed eigenvalue (1.151) was below its corresponding threshold (1.211).

Taken together, these findings supported the use of the seven-item composite as a study-specific dependency-related summary measure while recognizing the theoretical distinction between its two dimensions.

7.4 Correlation Analysis

Spearman's rank-order correlation analysis identified a statistically significant positive association between AI dependency and perceived negative problem-solving impact ($\rho = 0.387$, $p < .001$).

Withdrawal demonstrated a stronger positive correlation with the outcome ($\rho = 0.436$, $p < .001$), whereas Cognitive Preoccupation showed a weaker but statistically significant association ($\rho = 0.158$, $p = .017$).

Critical Thinking was weakly and negatively correlated with perceived negative problem-solving impact ($\rho = -0.144$, $p = .031$). Task Performance showed no statistically significant association ($\rho = -0.081$, $p = .226$).

Table 2. Spearman Correlations With Perceived Negative Problem-Solving Impact

Variable	Spearman's ρ	p-value
AI Dependency Score	0.387	< .001
Cognitive Preoccupation	0.158	.017
Withdrawal	0.436	< .001
Critical Thinking	−0.144	.031
Task Performance	−0.081	.226

Note. These correlations were exploratory, and no adjustment for multiple comparisons was applied. Statistical significance should therefore be interpreted cautiously.

7.5 Primary Ordinal Logistic Regression

The primary proportional-odds ordinal logistic regression model examined the association between the AI Dependency Score and perceived negative problem-solving impact while adjusting for country, academic year, and GenAI usage frequency.

The model converged successfully, with a log-likelihood of −333.007, Akaike Information Criterion (AIC) of 682.014, and Bayesian Information Criterion (BIC) of 709.413.

The AI Dependency Score was positively associated with higher categories of perceived negative problem-solving impact (adjusted OR = 2.946, 95% CI [2.065, 4.204], p < .001). This indicates that each one-point increase in the seven-item dependency score was associated with approximately 2.95 times the cumulative odds of reporting a higher category of perceived negative problem-solving impact, conditional on the included covariates and the proportional-odds model.

Country, academic year, and GenAI usage frequency were not statistically significant predictors at the conventional .05 threshold.

Table 3. Primary Adjusted Ordinal Logistic Regression Results

Predictor	Adjusted OR	95% CI	p-value
AI Dependency Score	2.946	2.065–4.204	< .001
Indonesia vs. United States	0.885	0.548–1.432	.620
Academic year	0.892	0.711–1.119	.322
AI usage frequency	0.735	0.518–1.042	.084

Note. OR = odds ratio; CI = confidence interval. Sample size = 227. The reported odds ratios assume a common predictor effect across cumulative outcome thresholds.

The findings provide evidence of an adjusted association between AI dependency-related experiences and students' perceptions of negative impacts on problem-solving. They do not establish that dependency caused changes in actual problem-solving performance.

7.6 Sensitivity and Robustness Analyses

We conducted several sensitivity analyses to evaluate the stability of the primary association. The unadjusted ordinal regression produced an odds ratio of 2.653 (95% CI [1.904, 3.695], p < .001). The primary adjusted model produced an odds ratio of 2.946, while an alternative specification using categorical covariate coding yielded an odds ratio of 2.797 (95% CI [1.955, 4.002], p < .001).

These results indicated that the positive association remained present across the examined specifications.

Dimension-specific analyses revealed that Cognitive Preoccupation was positively associated with perceived negative problem-solving impact when modeled separately (OR = 1.435, 95% CI [1.059, 1.944], p = .020). Withdrawal showed a stronger positive association (OR = 2.778, 95% CI [2.077, 3.716], p < .001).

When both dimensions were entered into the same adjusted model, Withdrawal remained statistically significant (OR = 2.891, 95% CI [2.105, 3.969], p < .001), whereas Cognitive Preoccupation was not statistically significant (OR = 0.897, 95% CI [0.639, 1.259], p = .529).

Table 4. Selected Sensitivity Analysis Results

Model specification	Dependency-related predictor	OR	95% CI
Unadjusted model	Seven-item composite	2.653	1.904–3.695
Primary adjusted model	Seven-item composite	2.946	2.065–4.204
Alternative covariate coding	Seven-item composite	2.797	1.955–4.002
Preoccupation-only model	Cognitive Preoccupation	1.435	1.059–1.944
Withdrawal-only model	Withdrawal	2.778	2.077–3.716
Joint-dimension model	Cognitive Preoccupation	0.897	0.639–1.259
Joint-dimension model	Withdrawal	2.891	2.105–3.969

Note. The dimension-specific models included the primary adjustment variables. Differences between odds ratios should not be interpreted as formal tests of differences in effect magnitude.

A leave-one-out influence analysis further showed that no single observation drove the primary dependency estimate. Across 227 refitted models, the estimated odds ratio ranged from 2.858 to 3.129, compared with 2.946 in the original model.

7.7 Cross-Country Comparison

An exploratory interaction analysis examined whether the association between AI dependency and perceived negative problem-solving impact differed between the United States and Indonesia.

The country-by-dependency interaction was not statistically significant (interaction OR = 0.841, 95% CI [0.443, 1.597], p = .596). A likelihood-ratio comparison produced a similar result ($\chi^2 = 0.281$, p = .596).

Model-derived country-specific associations were positive in both groups. The estimated odds ratio was 3.206 (95% CI [1.993, 5.158]) for respondents from the United States and 2.696 (95% CI [1.663, 4.370]) for respondents from Indonesia.

These findings did not provide statistically significant evidence that the association differed between the two national groups. However, the absence of a statistically significant interaction should not be interpreted as proof that the associations were identical. Furthermore, the analysis did not establish cross-cultural measurement equivalence, and the country comparisons should be interpreted as exploratory.

7.8 Model Diagnostics

Variance inflation factors ranged from 1.007 to 1.133, indicating no substantial linear multicollinearity among the primary predictors.

The ordinal regression model also converged successfully, supporting the numerical stability of its estimation.

Threshold-specific binary logistic regressions produced dependency-related odds ratios ranging from approximately 2.326 to 3.293. These estimates were broadly positive across outcome thresholds, although their variation suggested that the proportional-odds assumption warranted caution.

A multinomial logistic regression was fitted as an alternative specification. The ordinal model had lower AIC and BIC values than the multinomial model, indicating more favorable penalized fit under those criteria.

Nevertheless, these comparisons did not constitute a definitive formal test of the proportional-odds assumption. Consequently, the primary findings should be interpreted conditional on the specified ordinal model.

7.9 Summary of Findings

The analysis identified a consistent positive association between generative AI dependency-related experiences and students' perceived negative effects of AI use on problem-solving skills or efficiency.

The primary adjusted model indicated that higher dependency scores were associated with greater odds of reporting a more negative perceived problem-solving impact. Sensitivity analyses supported the stability of this association across several model specifications. Withdrawal-related experiences demonstrated a particularly consistent association with the outcome, including when Cognitive Preoccupation was included in the same model.

Exploratory country comparisons did not identify statistically significant evidence of different associations between respondents from the United States and Indonesia. Overall, the findings suggest that students reporting stronger AI dependency-related experiences were also more likely to perceive negative effects of GenAI use on their problem-solving skills or efficiency. These results remain observational and do not establish causal relationships or objectively measured cognitive decline.

8. Discussion

8.1 Overview of the Main Findings

This study examined the association between generative artificial intelligence (GenAI) dependency and university students' perceived negative effects of AI use on their problem-solving skills or efficiency. Using survey data from 227 active GenAI users in the United States and Indonesia, the analysis identified a statistically significant positive association between dependency-related experiences and perceived negative problem-solving impact. The primary adjusted ordinal logistic regression indicated that a one-point increase in the seven-item AI Dependency Score was associated with approximately 2.95 times the cumulative odds of reporting a higher category of perceived negative problem-solving impact (adjusted OR = 2.946, 95% CI [2.065, 4.204], $p < .001$).

This association remained positive across alternative model specifications and leave-one-out sensitivity analyses, suggesting that the finding was not dependent on a single observation or the particular covariate specification examined.

However, the result should be interpreted as an association between self-reported experiences rather than evidence that reliance on GenAI causes deterioration in students' actual problem-solving abilities.

8.2 AI Dependency and Perceived Problem-Solving Impact

The positive association observed in this study suggests that students who reported stronger dependency-related experiences were also more likely to agree that their use of GenAI had negatively affected their problem-solving skills or efficiency.

This finding is relevant to existing concerns about the relationship between AI-assisted learning and independent cognitive engagement. Although GenAI can provide explanations, generate solutions, and support academic activities, excessive psychological reliance may

coexist with students' perceptions that their independent problem-solving capabilities are being affected.

The finding should not be interpreted as evidence that AI assistance is inherently harmful. Previous research has identified potential educational benefits associated with generative AI, including improved access to learning resources, personalized explanations, and support for higher-order learning activities (Kasneci et al., 2023; Chen & Cheung, 2025).

The distinction between productive AI assistance and dependency-related experiences is therefore important. Students may benefit from using AI to clarify concepts or evaluate alternative solutions while still maintaining responsibility for independent reasoning and decision-making.

The present study contributes to this discussion by examining dependency-related experiences separately from the questionnaire item used to measure perceived negative problem-solving impact. Although this approach reduces direct measurement overlap, both variables originate from the same self-report instrument, and common-method influences remain possible.

8.3 Cognitive Preoccupation and Withdrawal

One of the notable findings was the difference between the Cognitive Preoccupation and Withdrawal dimensions.

When examined separately, both dimensions demonstrated statistically significant positive associations with perceived negative problem-solving impact. However, when both were included in the same adjusted model, Withdrawal remained statistically significant, whereas Cognitive Preoccupation did not.

Withdrawal-related items describe experiences such as restlessness, distraction, frustration, and discomfort when GenAI tools are unavailable. The observed association suggests that these experiences may be particularly relevant to students' perceptions of difficulties with independent problem-solving.

In contrast, Cognitive Preoccupation reflects frequent thoughts about AI use and the desire to engage with AI even when it is unnecessary. Although this dimension was associated with the outcome in its separate model, its adjusted association was not statistically significant after accounting for Withdrawal.

This pattern suggests that the relationship between dependency-related experiences and perceived negative problem-solving impact may not be uniform across dimensions.

Nevertheless, the analyses do not establish that Withdrawal is causally responsible for perceived difficulties, nor do they demonstrate that its association is statistically stronger than Cognitive Preoccupation through a formal comparison of coefficients.

The findings instead support further investigation into the potentially distinct roles of dependency dimensions in students' academic experiences.

8.4 Cross-Country Findings

The study also examined whether the association between AI dependency and perceived negative problem-solving impact differed between respondents from the United States and Indonesia.

The interaction analysis did not identify a statistically significant difference between the two groups ($p = .596$). Model-derived associations were positive in both the United States and Indonesia, with overlapping confidence intervals.

These findings suggest that the positive association was observable within both national groups represented in the dataset. However, the absence of a statistically significant interaction does not establish that the relationships are identical.

Cross-cultural interpretation also requires caution because differences in educational practices, technological environments, questionnaire interpretation, and response styles may influence observed results.

Furthermore, the study did not formally establish measurement equivalence across the two national groups. Consequently, the findings should not be interpreted as definitive evidence that AI dependency operates similarly across cultures.

Future research using larger and more representative samples, alongside formal measurement-invariance testing, would provide a stronger basis for cross-cultural comparison.

8.5 Measurement and Methodological Implications

The study's measurement strategy addressed an important challenge in examining the relationship between AI dependency and its perceived consequences.

The original dependency questionnaire included a Negative Consequences dimension containing the problem-solving outcome item. Including that item within the dependency predictor would have created direct item overlap and potentially inflated the estimated association.

By constructing the primary dependency measure from Cognitive Preoccupation and Withdrawal items only, the study reduced this direct overlap.

The seven-item composite demonstrated good internal consistency (Cronbach's $\alpha = .801$). Exploratory factor analysis also identified meaningful shared variation among the items. However, the dimensionality results require careful interpretation. A specified two-factor solution separated Cognitive Preoccupation and Withdrawal items, whereas parallel analysis suggested retaining one dominant factor. These findings support continued investigation of the composite's structure rather than establishing its validity as a newly standardized instrument.

The use of ordinal logistic regression was appropriate for preserving the ordered nature of the five-category outcome. Nevertheless, threshold-specific diagnostics showed some variation in estimated associations, and the proportional-odds assumption was not definitively established through a formal test.

Accordingly, the primary odds ratio should be interpreted as conditional on the specified proportional-odds model.

8.6 Implications for Higher Education

The findings have potential implications for how universities approach the integration of GenAI into teaching and learning.

First, educational guidance may benefit from distinguishing ordinary AI use from dependency-related experiences. Frequent use alone should not be treated as evidence of harmful reliance.

Second, universities could encourage students to use GenAI as a supplementary learning resource rather than a substitute for independent reasoning. Activities that require students to explain their reasoning, evaluate AI-generated responses, and solve problems without automated assistance may help preserve opportunities for active cognitive engagement.

Third, the observed association involving Withdrawal suggests that future educational research should consider students' psychological experiences when AI tools are unavailable, rather than focusing exclusively on usage frequency.

These recommendations are practical considerations arising from the observed associations and broader educational literature. The present study did not evaluate interventions and therefore cannot establish whether such strategies improve problem-solving outcomes.

8.7 Overall Interpretation

The findings provide evidence that stronger self-reported GenAI dependency-related experiences are associated with greater perceived negative effects on problem-solving skills or efficiency among the students included in this dataset.

The association was consistent across several statistical specifications, and Withdrawal-related experiences showed a particularly notable relationship with the outcome.

At the same time, the cross-sectional design, self-reported measurements, and limitations of the modified dependency composite restrict causal and generalizable interpretations.

The study therefore contributes to understanding how students perceive the relationship between their reliance on GenAI and their problem-solving experiences, while highlighting the need for longitudinal research and objective cognitive assessments.

9. Limitations and Future Research

9.1 Cross-Sectional Research Design

The primary limitation of this study is its cross-sectional research design. Because all variables were measured at a single point in time, the analysis cannot establish temporal relationships or causal effects between generative artificial intelligence (GenAI) dependency and perceived negative problem-solving impact.

Although higher dependency scores were associated with greater odds of reporting negative problem-solving experiences, it remains unclear whether dependency-related experiences preceded these perceptions or whether students who already experienced difficulties with independent problem-solving were more likely to develop stronger reliance on GenAI.

Future research should employ longitudinal designs to examine changes in dependency-related experiences and problem-solving outcomes over time. Experimental or intervention-based studies could also investigate whether different patterns of AI assistance influence students' independent cognitive performance.

9.2 Reliance on Self-Reported Measurements

The study relied on self-reported questionnaire responses, which may be influenced by recall errors, social desirability, individual expectations, and differences in how participants interpret survey statements.

Importantly, the dependent variable measured students' agreement that GenAI use had negatively affected their problem-solving skills or efficiency. It did not assess problem-

solving ability through standardized cognitive tests, academic performance records, or direct observation.

Consequently, the findings cannot establish an actual decline in students' problem-solving capabilities.

Future studies should incorporate objective assessments, including structured problem-solving tasks, independent reasoning exercises, and performance-based evaluations. Combining subjective perceptions with objective measures would provide a more comprehensive understanding of the relationship between GenAI use and cognitive functioning.

9.3 Measurement Validity and Common-Method Bias

The primary dependency measure was constructed from seven selected items of the original Generative AI Dependency Scale. The Negative Consequences dimension was excluded to reduce direct overlap between the predictor and the problem-solving outcome.

Although the resulting composite demonstrated good internal consistency (Cronbach's α = .801), reliability alone does not establish construct validity.

Exploratory factor analysis revealed a recognizable two-dimensional pattern corresponding to Cognitive Preoccupation and Withdrawal, whereas parallel analysis suggested retaining one dominant factor. These findings indicate that the dimensional structure requires further investigation.

Additionally, the predictor and outcome originated from the same questionnaire, creating the possibility of common-method bias and conceptual overlap despite the exclusion of directly overlapping items.

Future research should validate the modified composite using independent samples, ordinal-data-appropriate psychometric techniques, and confirmatory factor analysis. Where possible, researchers should measure dependency and problem-solving outcomes using different instruments or assessment methods.

9.4 Sample Characteristics and Generalizability

The analytical sample consisted of 227 active GenAI users drawn from a dataset of business students in the United States and Indonesia.

Although the inclusion of respondents from two countries provided an opportunity for exploratory comparison, the sample cannot be assumed to represent all university students in either country or globally.

The study also excluded respondents who did not meet the active-user criterion, limiting its applicability to students who do not use GenAI.

Furthermore, the sample size and composition restricted the precision of subgroup analyses, particularly when examining potential differences between national groups.

Future studies should include larger, more diverse samples across academic disciplines, institutions, countries, and levels of GenAI engagement. Probability-based sampling, where feasible, could improve population-level generalizability.

9.5 Statistical Model Assumptions

The primary analysis employed proportional-odds ordinal logistic regression to model the five-category outcome.

Although the model converged successfully and the dependency association remained positive across several sensitivity analyses, the proportional-odds assumption was not definitively established through a formal diagnostic.

Threshold-specific models indicated some variation in predictor estimates across outcome categories. Therefore, the reported common odds ratio should be interpreted in relation to the assumptions of the fitted ordinal model.

Future analyses could consider formal proportional-odds diagnostics, partial proportional-odds models, and alternative ordinal modeling approaches to evaluate whether the association varies across response thresholds.

The analysis also involved multiple exploratory comparisons without a formal multiplicity adjustment, increasing the possibility of chance findings among secondary outcomes.

9.6 Residual Confounding and Cross-Cultural Measurement

The regression models adjusted for country, academic year, and frequency of GenAI use. However, potentially relevant characteristics such as academic achievement, prior problem-solving ability, motivation, educational practices, and psychological factors were not comprehensively controlled.

Residual confounding may therefore influence the observed associations.

In addition, the study did not establish measurement invariance between respondents from the United States and Indonesia. Differences in language, cultural interpretation, or response styles could affect comparisons between the two groups.

Future cross-cultural research should evaluate measurement equivalence before drawing strong conclusions about similarities or differences between national populations.

9.7 Directions for Future Research

Future investigations should prioritize longitudinal and experimental designs, objective problem-solving assessments, independent validation of dependency measures, and more representative student samples.

Further research should also examine whether Cognitive Preoccupation and Withdrawal have distinct relationships with learning behaviors and academic outcomes.

Finally, intervention studies could evaluate educational strategies that encourage responsible GenAI use while preserving opportunities for independent reasoning, critical evaluation, and problem-solving practice.

10. Conclusion

This study investigated the association between generative artificial intelligence dependency and university students' perceived negative effects of AI use on their problem-solving skills or efficiency. Using secondary survey data from 227 active GenAI users in the United States and Indonesia, the research employed descriptive statistics, reliability analysis, correlation analysis, ordinal logistic regression, and sensitivity analyses.

The primary adjusted ordinal regression identified a statistically significant positive association between the seven-item AI Dependency Score and perceived negative problem-solving impact (adjusted OR = 2.946, 95% CI [2.065, 4.204], $p < .001$). The association remained positive across alternative model specifications and leave-one-out analyses.

Dimension-specific analyses indicated that Withdrawal-related experiences were consistently associated with the outcome, including when Cognitive Preoccupation was included in the same model. Exploratory cross-country analyses did not provide statistically significant evidence that the primary association differed between respondents from the United States and Indonesia.

The study also demonstrated the importance of measurement design by excluding directly overlapping Negative Consequences items from the dependency predictor. Although the resulting seven-item composite demonstrated good internal consistency, further psychometric validation is necessary.

Overall, the findings suggest that stronger GenAI dependency-related experiences are associated with greater perceived negative effects on problem-solving among the students examined. However, the cross-sectional design and reliance on self-reported measurements prevent conclusions about causation or objectively measured cognitive decline.

The research contributes to ongoing discussions about responsible AI integration in higher education and highlights the importance of distinguishing beneficial AI assistance from dependency-related experiences. Future research using longitudinal designs, validated instruments, and objective assessments will be necessary to determine how patterns of GenAI reliance relate to students' actual problem-solving performance.

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